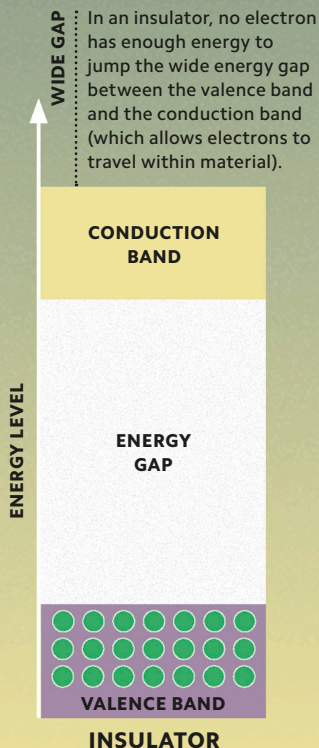
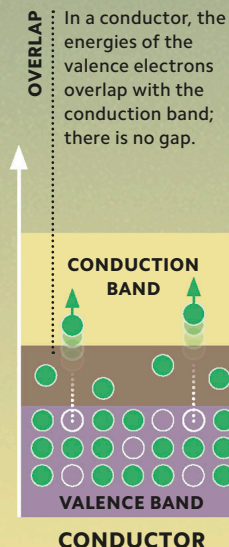
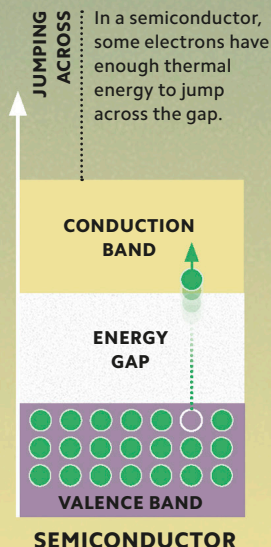




Allowed energy states

The ranges of energies where no possible states can exist in a solid are known as band, or energy, gaps. Every solid has a characteristic structure of bands and gaps. The valence band contains the highest energy level that electrons occupy.



SOLID STATES

In order to understand the electric properties of materials, it is necessary to understand how electrons move within them. Band theory is a model that describes the permitted and forbidden energy levels—bands and band gaps—in materials with solid structures (see pp.76–77), which restrict the behavior of electrons. This model explains thermal and electric properties of solid matter and is the basis of solid-state electronic devices, such as transistors (see opposite), diodes (see p.92), and solid-state data storage devices.